DSP - LEC - 11 – CH – 4- Z-Tr. App.s

CHAPTER 4

Z-TRANSFORM APPLICATIONS IN DSP

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4.5 SOME APPLICATIONS OF THE Z-TRANSFORM IN DSP

4.5.1 Pole-zero description of discrete-time systems

In most practical discrete-time systems the z-transform, that is the system transfer function H(z), can be expressed in terms of its poles and zeros. Consider for example, the following z-transform representing a general, N^{th}-order discrete-time filter (where N = M):

$$H(z) = \frac{N(z)}{D(z)} \qquad (4.34)$$

Where :

$$N(z) = a_0 z^N + a_1 z^{N-1} + a_2 z^{N-2} + \ldots\ldots + a_N$$

$$D(z) = b_0 z^N + b_1 z^{N-1} + b_2 z^{N-2} + \ldots\ldots + b_N$$

The a_k and b_k are the coefficients of the filter.

If H(z) has poles at $z = p_1, p_2, \ldots\ldots\ldots, p_N$ and zeros at $z = z_1, z_2, \ldots\ldots, z_N$, then H (z) can be factored and represented as

$$H(z) = \frac{K(z - z_1)(z - z_2).....(z - z_N)}{(z - p_1)(z - p_2).....(z - z_N)} \qquad (4.35)$$

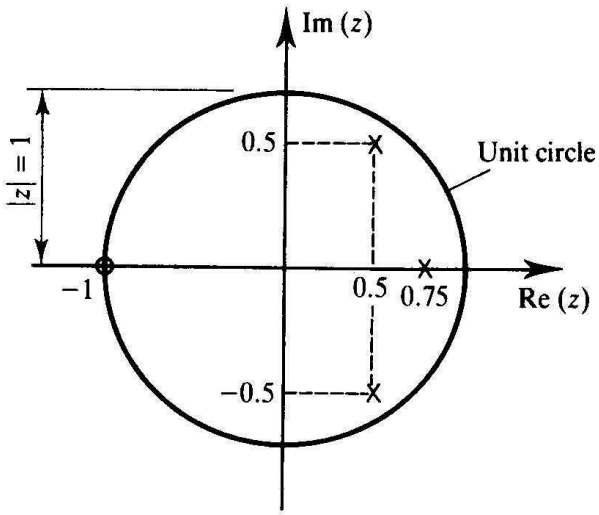


Fig. 4.6 Description of a z-transform in the form of a pole-zero diagram: X, pole; o, zero.

Where z_i is the ith zero, p_i is the ith pole and K is the gain factor. You will recall that the poles of a z-transform such as H(z) are the values of z for which H(z) becomes infinity. The values of z for which H(z) becomes zero are referred to as zeros. The poles and zeros of H(z) may be real or complex. When they are complex, they occur in conjugate pairs, to ensure that the coefficients, a_k and b_k, are real. It should be clear from Eq. 4.35 that if the locations of the poles and zeros of H(z) are known, then H(z) itself can be readily reconstructed to within a constant.

The information contained in the z-transform can be conveniently displayed as a pole-zero diagram; see for example Fig. 4.6. In the diagram, X marks the position of a pole a o denotes the position of a zero. For this example, the poles are located at z = 0.5 $\pm$ 0.5j, and at z = 0.75. A single zero is at z = -1. An important feature of the pole-zero diagram is the unit circle, that is the circle defined by |z| = 1; see Fig. 4.6. As will become clear, the unit circle plays an important role in the analysis and design of discrete-time systems.

The pole-zero diagram provides an insight into the properties of a given discrete-time system. For example, from the locations of the poles and zeros we can infer the

frequency response of the system as well as its degree of stability. For a stable system, all the poles must lie inside the unit circle (or be coincident with zeros on the unit circle).

Often, the z-transform is not available in factored form but as a ratio of polynomials such as Eq. 4.34. In these cases, describing the z-transform, H(z), in terms of its poles and zeros will require finding the roots of the denominator polynomial, D(z), and those of the numerator polynomial, N(z).

For a second-order polynomial which has the form

$$ax^2+bx+c,$$

the roots are given by

$$\frac{-b \pm (b^2 - 4ac)^{1/2}}{2a} \qquad \textbf{(4.36)}$$

For higher-order polynomials, finding the roots of N(z) or D(z) is a difficult task. In practice, this is often achieved using numerical methods involving, for example, Newton's and/or Baistow's algorithms (see for example, Atkinson and Harley (1983)). The need to find the poles and zeros often arises in connection with the design of discrete filters and in stability analysis. Fortunately, in the case of discrete-time filter design the poles and zeros are automatically generated by the filter design software, obviating the need to find the roots directly.

Example 4.10

(1) Express the following transfer function in terms of its poles and zeros and sketch the pole-zero diagram:

$$H(z) = \frac{1 - z^{-1} - 2z^{-2}}{1 - 1.75z^{-1} + 1.25z^{-2} - 0.375z^{-3}}$$

(2) Determine the transfer function, H(z), of a discrete-time filter with the pole-zero diagram shown in Fig. 4.7.

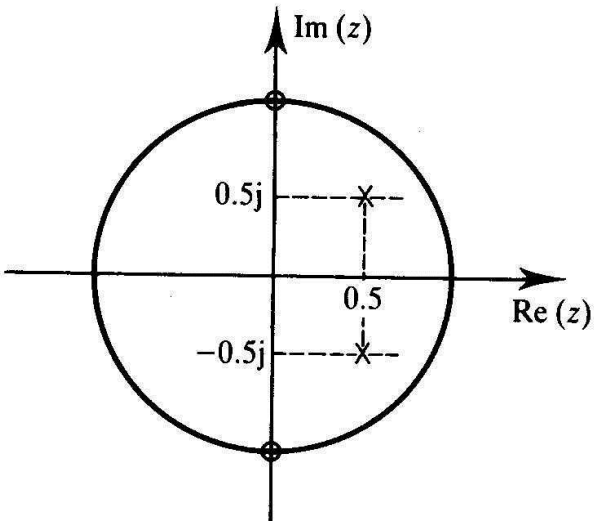


Fig. 4.7 Pole-zero diagram for Example (4.10), part (2)

Solution

(1) First, we express H(z) positive powers of z and then factorize it so that its poles and zeros can be determined. If we multiply the denominator and numerator by z^3, the highest power of z, we obtain.

$$H(z) = \frac{z(z^2 - z - 2)}{z^3 - 1.75z^2 + 1.25z - 0.375}$$

Factoring, we have

$$H(z) = \frac{z(z-2)(z+1)}{(z-0.5+j0.5)(z-0.5-j0.5)(z-0.75)}$$

Thus, the pole locations are at z = 0.5 $\pm$ 0.5j and at z = 0.75. The zeros are at z = 0, z=2 and z= -1. The pole-zero diagram is depicted in Fig. 4.8.

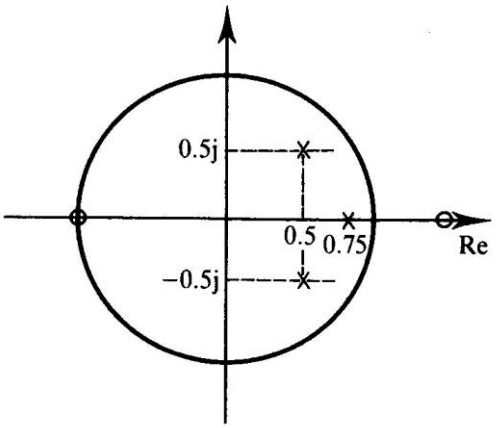


Fig. 4.8 Pole-zero diagram for Example (4-10), part (1).

(2) From the pole-zero diagram of Fig. 4-7, the zeros of the transfer function are at z = $\pm$j and the poles are at z = 0.5 $\pm$ 0.5j. The transfer function can be written down directly:

$$H(z) = \frac{K(z-j)(z+j)}{(z-0.5-0.5j)(z-0.5+0.5j)}$$

$$= \frac{K(z^2+1)}{z^2-z+0.5}$$

$$= \frac{K(1+z^{-2})}{1-z^{-2}+0.5z^{-2}}$$

4.5.2 Stability Considerations of discrete-time systems

Stability analysis is often carried out as part of the design of discrete-time systems. A useful stability criterion for LTI systems is that all bounded inputs produce bounded outputs. This is the so-called BIBO (bounded input, bounded output) condition. An LTI system is said to be BIBO stable if and only if it satisfies the criterion.

$$\sum_{k=0}^{\infty} |h(k)| < \infty \qquad (4.37)$$

where h(k) is the impulse response of the system. It is obvious that if the impulse response is of finite length the condition above is satisfied since the sum of the impulse response coefficients will be finite. Thus, stability considerations apply only to systems with impulse response of infinite duration.

For the output to be bounded all the poles must lie inside the unit circle. When a pole lies outside the unit circle the system is unstable. In practice, a system with a pole on the unit circle is also regarded as unstable or potentially unstable, since a minor disturbance or error will invariably push the system into instability. An exception is when a pole is coincident with a zero on the unit circle so that its effects are nullified. For an unstable system, the impulse response will increase indefinitely with time.

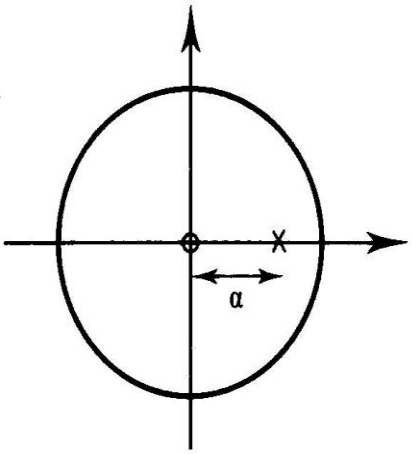
In principle, testing for stability is simple: find the pole positions of the z-transform. If any pole is on or outside the unit circle (unless it is coincident with a zero on the unit circle) then the system is unstable. In practice, finding the positions of the poles may not be an easy task.

A simple test that may be used when the system z-transform, H(z), is not available in factored form is to obtain and plot a sufficient number of values of the impulse response, by finding the inverse z-transform. If the impulse response increases indefinitely with time or fails to decay fast enough then the system is either unstable or marginally stable.

Figure 4.9 shows examples of the behaviour of the impulse response of a simple discrete-time system for various degrees of stability.

	$h(n)$			
n	α=0.5	α=0.99	α=1	α=1.5
0	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00
1	1.00000e+01	1.00000e+01	1.00000e+01	1.00000e+01
2	5.00000e+00	9.90000e+00	1.00000e+01	1.50000e+01
3	2.50000e+00	9.80100e+00	1.00000e+01	2.25000e+01
4	1.25000e+00	9.70299e+00	1.00000e+01	3.37500e+01
5	6.25000e−01	9.60596e+00	1.00000e+01	5.06250e+01
6	3.12500e−01	9.50990e+00	1.00000e+01	7.59375e+01
7	1.56250e−01	9.41480e+00	1.00000e+01	1.13906e+02
8	7.81250e−02	9.32065e+00	1.00000e+01	1.70859e+02
9	3.90625e−02	9.22745e+00	1.00000e+01	2.56289e+02

(a)



(b)

Fig. 4.9 An illustration of the behavior of the impulse response of a system for various degrees of stability: (a) impulse response; (b)z-plane pole-zero diagram.

The system z-transform is $10z^{-1}/(1-\alpha z^{-1})$.

(i) For α = 0.5 system is stable,

(ii) for α =0.99 it is marginally stable,

(iii) for α =1 it is potentially unstable, and

(iv) for α =1.5 it is unstable.

Notice, for example, that for α =0.5 the impulse response values decay rapidly as n increases, while, for α =1.5, the impulse response values increase rapidly.

4.5.3 Difference equations of discrete-time systems

The difference equation specifies the actual operations that must be performed by the discrete-time system on the input data, in the time domain, in order to generate the

desired output. The difference equation, for most practical cases of interest, may be written as

$$y(n) = \sum_{k=0}^{N} a_k x(n-k) - \sum_{k=1}^{M} b_k y(n-k) \qquad (4.38)$$

where x(n) is the input sample, y(n) is the output sample, y(n–k) is the previous output and a_k, b_k are system coefficients. Equation 4-38 indicates that the current output, y(n), is obtained from present and past input samples and previous outputs, y(n-k).

The difference equations for discrete-time systems are readily obtained from their transfer functions and vice versa using the delay property of the z-transform :

$$a_k x(n) \leftrightarrow a_k X(z)$$

$$a_k x(n-k) \leftrightarrow a_k z^{-k} X(z)$$

Thus, Eq. 4.38 may be written as

$$Y(z) = \sum_{k=0}^{N} a_k z^{-k} X(z) - \sum_{k=1}^{M} b_k z^{-k} Y(z) \qquad (4.39)$$

Simplifying, we obtain the z-domain transfer function of the discrete system, H(z) :

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{k=0}^{N} a_k z^{-k} \; / \; (1 + \sum_{k=1}^{M} b_k z^{-k}) \qquad (4.40)$$

If all the denominator coefficients, b_k, are zero. Eqs. 4.38 and 4.39 reduce to

$$y(n) = \sum_{k=0}^{N} a_k x(n-k) \qquad (4.41a)$$

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{k=0}^{N} a_k z^{-k} \qquad (4.41b)$$

The output of the system, y(n), now depends only on the present and past input samples but not on previous outputs as in Eq. 4.38. The coefficients, the a_k, in this case represent the impulse response of the system and are normally denoted by the symbol h(k). This class of LTI system is referred to as finite impulse response (FIR) systems, since the length of h(k) is clearly finite.

Systems characterized by Eqs. 4.38 and 4.40 where at least one of the denominator coefficients is nonzero are referred to as infinite impulse response (IIR) systems. In IIR systems at least one of the poles will be non-zero, but FIR systems normally contain no poles.

4.5.4 Impulse response estimation of discrete-time systems

In the design of discrete-time systems the need often arises to obtain values of impulse responses. For example, in FIR system design the impulse response is required to implement the system, and in IIR system design the values are required for stability analysis. The impulse response may also be used to evaluate the frequency response of the system.

The impulse response of a discrete-time system may be defined as the inverse z-transform of the system's transfer function, H(z):

$$h(k) = Z^{-1}[H(z)],\ k = 0, 1, \ldots\ldots..$$

If the z-transform, H(z), is available as a power series, that is

$$H(z) = \sum_{n=0}^{\infty} h(n)z^{-n}$$

$$= h(0) + h(1)z^{-1} + h(2)z^{-2} + h(3)z^{-3} + \ldots.. \qquad \textbf{(4.42)}$$

the coefficients of the z-transform give directly the impulse response. For IIR systems, H(z) is often expressed as a ratio of polynomials such as Eq. 4.9. In this case, the IZT

methods described in Section 4.2 can be used to obtain the impulse response of the system.

$$H\left(e^{j\omega T}\right)= \left.\frac{a_0 + a_1 z^{-1} \ldots + a_N z^{-N}}{b_0 + b_1 z^{-1} + \ldots + b_M z^{-M}}\right|_{z=e^{j\omega T}} \qquad (4.9)$$

The impulse response may also be viewed as the response of a discrete-time system to a unit impulse, u(n), which has a value of 1 at n=0 and a value of 0 at all other values of n. This view arises from the fact that if we make the input to the system equal to the unit impulse, x(n) = u(n), the output of the system is in fact equal to h(n), the system's impulse response :

$$y(n) = \sum_{n=0}^{\infty} h(k)x(n-k) = \sum_{n=0}^{\infty} h(k)u(n-k)$$

$$= h(0)u(n)+h(1)u(n-1) + h(2)u(n-2) + \ldots\ldots$$

$$= h(n),\ \ n = 0, 1, \ldots\ldots\ldots \qquad (4.43)$$

This provides a simple alternative method of computing h(n) (indeed it provides another method of obtaining the inverse z-transform) as illustrated by the following example.

Example 4.11

Find the impulse response of the discrete-time filter characterized by the following z-transfer function:

(1) by using the power series method and

(2) by applying a unit impulse to the system

$$H(z) = \frac{1-z^{-1}}{1+0.5z^{-1}}$$

Solution

(1) Using the power series method, the values of the impulse response are obtained as

$$1-1.5z^{-1}+0.75z^{-2}-0.325z^{-3}.....$$

$$1+0.5z^{-1}\,\big|\,1-z^{-1}$$

$$1+0.5z^{-1}$$

$$-1.5z^{-1}$$

$$-1.5z^{-1}-0.75z^{-2}$$

$$0.75\,z^{-2}$$

$$0.75z^{-2}+0.375z^{-3}$$

$$-\,0.3725z^{-3}$$

From the quotients the impulse response values are

$$h(0) = 1,\ h(1) = -\,1.5,\ h(2) = 0.75,\ h(3) = -0.325$$

(2) First, we need to obtain the difference equation of the filter from the transfer function:

$$H(z)=\frac{Y(z)}{X(z)}=\frac{1-z^{-1}}{1+0.5z^{-1}}$$

Cross-multiplying and using the delay property of the z-transform we obtain the difference equation:

$$Y(z) + 0.5Y(z)z^{-1} = X(z) - X(z)z^{-1}$$

$$y(n) + 0.5y(n-1) = x(n) - x(n-1)$$

Simplifying we have

$$y(n) = x(n) – x(n – 1) – 0.5y (n – 1)$$

The impulse response of the filter can now be obtained by letting x(n) = u(n) where

$$u(n) = 1, \quad n = 0$$
$$= 0, \quad n \neq 0$$

and assuming an initial condition of y(– 1) = 0 :

$$y(0) = 1$$

$$y(1) = x(1) – x(0) – 0.5y(0) = 0 – 1 – 0.5 = - 1.05$$

$$y(2) = x(2) – x(1) – 0.5y(1) = -0.5 \times -1.05 = 0.75$$

$$y(3) = x(3) – x(2) – 0.5y(2) = -0.5 \times 0.75 = -0.325$$

From this the impulse response values are

$$h(0) = 1, h(1) = - 1.5, h(2) = 0.75, h(3) = - 0.375$$

It is seen that both method lead to identical results.

4.5.5 Realization structures for digital filters

Discrete-time filters are often represented in the form of block or signal flow diagrams. The diagrams are a convenient way of representing the difference equations or equivalently the transfer functions. Consider for example, a simple discrete filter with the following difference equation:

$$y(n) = x(n-1) - b_1 y(n-1) + b_2 y(n-2) + b_3 y(n-3) \qquad (4.44)$$

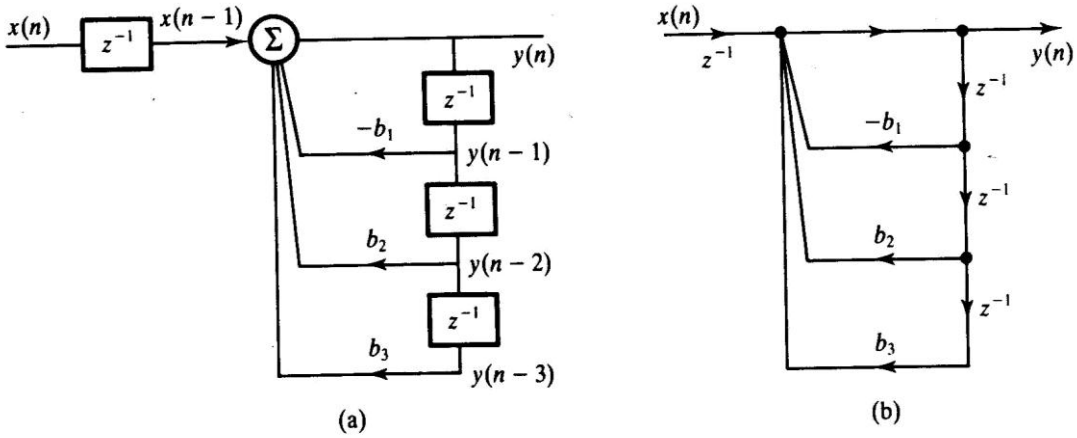


Fig. 4.10 Realization diagrams for a difference equation:

(a) in block diagram form; (b) signal flow diagram.

The block diagram representation of this equation is shown in Fig. 4.10(a). In the figure, the symbol z^{-1} represents a delay of 1 unit of time as may be deduced from the signals at various nodes; the arrows represent multipliers and the constants next to them the multiplication factors. The relationship between the difference equation and the block diagram should be evident. A signal flow diagram representation of the same difference equation is shown in Fig. 410(b). It is common to refer to the block or flow diagram as a realization diagram.

When the order of H(z) is high, the discrete-time filter is rarely realized directly as Fig. 4.10 implies because large errors will result if a small number of bits is used to represent the coefficients and to execute the difference equations. The common practice

is to decompose the transfer function into a cascade or parallel combination of second- and/or first-order z-transforms. For cascade realization, the transfer function, H(z), is factored as

$$H(z) = H_1(z)\, H_2(z) \ldots H_k(z) = \prod_{i=1}^{k} H_i(z) \qquad \textbf{(4.45)}$$

where $H_i(z)$ is either a second- or a first-order section:

$$H_i(z) = \frac{a_0 + a_{1i} z^{-1} + a_{2i} z^{-2}}{1 + b_{1i} z^{-1} + b_{2i} z^{-2}} \qquad \text{second order}$$

$$H_i(z) = \frac{a_0 + a_{1i} z^{-1}}{1 + b_{1i} z^{-1}} \qquad \text{first order}$$

and K is the integer part of (M + 1)/2. The overall z-transform is the product of the individual z-transforms: see Fig. 4.11.

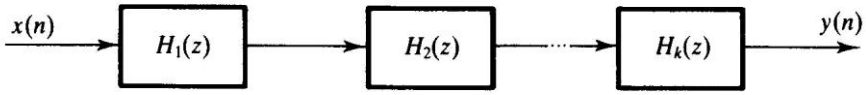


Fig. 4.11 General structure for cascade realization.

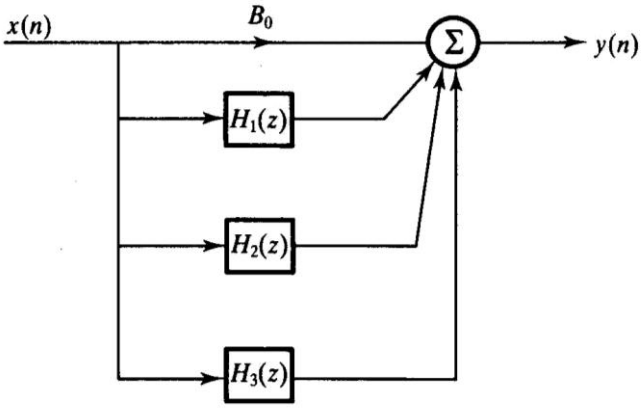


Fig. 4.12 General structure for parallel realization.

For parallel realization, the transfer function is decomposed, using partial fractions, to give

$$H(z) = B_0 + \sum_{i=1}^{k} H_i(z) \qquad \textbf{(4.46)}$$

where, as above, $H_i(z)$ is either a second-order or a first-order section but of the form

$$H_i(z) = \frac{a_{0i} + a_{1i}z^{-1}}{1 + b_{1i}z^{-1} + b_{2i}z^{-2}} \qquad \text{second order}$$

$$H_i(z) = \frac{a_0}{1 + b_{1i}z^{-1}} \qquad \text{first order}$$

where K is the integer part of (M+1)/2 and

$$B_0 = a_N / b_M$$

Figure 4.12 shows the general structure for the parallel realization.

In the design of digital filters, software packages are normally used to obtain the coefficients, a_{ki} and b_{ki}, above. Unfortunately, most software packages produce coefficients only for the cascade structure. The coefficients for the parallel structure can be obtained from those of the cascade structure by using a partial fraction expansion.